QFT, The Standard Model, and Supersymmety

Chance Emrich

May 26, 2026

Part N

Contents

Part I

Math Toolkit

1.1. Notation (Read everything or you won't be able to understand later, or come back to it later)

- (Loose) Einstein Summation notation is used in which if an index appears twice in both top and bottom, such as $\partial_\mu \partial^\mu$, it is equivalent to summing over said indices such as $\sum_{\mu \in \{0,1,2,3\}} \partial_\mu \partial^\mu$
  - The reason that this is loose is due to the fact that if two indices appear twice in an expression but they are both on the bottom, it is implied that the reader should raise the second appearance of the index. The following is an example so the reader doesn't get confused: $\partial_\mu \partial_\mu = \partial_\mu \partial^\mu = \sum_{\mu \in \{0,1,2,3\}} \partial_\mu \partial^\mu$
- The *Dirac Delta Function* is defined as

$$\delta(x) = \begin{cases} 0 & \text{if } x \neq 0 \\ \infty & \text{if } x = 0 \end{cases} \tag{1}$$

with

$$\int dx \ \delta(x) = 1 \tag{2}$$

This is not, in fact, a function, due to its unorthodox definition. Instead, it is a distribution (it can be defined as the limit of several different distributions, such as the *normal distribution*, a *uniform distribution*, etc.).

- $\vec{x} = x^i$, $i \in \{1,2,3\}$ (spatial components of x)
- The *d'Alembertian* operator $\square \equiv \partial_\mu \partial^\mu$

This operator is useful in *Lagrangians* and the *Klein-Gordon equation.*

- The *Laplacian* operator $\triangle \equiv \partial_i \partial^i$ where $i \in \{1,2,3\}$
- Natural Units are used in which $\hbar = c \ = 1$
- Both Sided Derivative: $\phi^* \overleftrightarrow{\partial_\mu} \phi \equiv \left(\phi^* \partial_\mu \phi - \phi \partial_\mu \phi^*\right)$
- Feynman Slash: $\not{p} \equiv p_\mu \gamma^\mu$
- Pauli matrices

$$\sigma^0 = \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}, \sigma^1 = \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}, \sigma^2 = \begin{pmatrix} & -i \\ i & \end{pmatrix}, \sigma^3 = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix} \tag{3}$$

and $\sigma^\mu \equiv \left(\sigma^0, \vec{\sigma}\right)$, $\bar{\sigma}^\mu \equiv \left(\sigma^0, -\vec{\sigma}\right)$

- $$\gamma^\mu = \begin{pmatrix} & \sigma^\mu \\ \bar{\sigma}^\mu & \end{pmatrix} \tag{4}$$

  in the Weyl representation

- $\bar{\psi} \equiv \psi^\dagger \gamma^0$ where ψ is a Dirac spinor
- Left Handed Derivative

$$\bar{\psi}\left(-i\overleftarrow{\not\partial}-m\right)$$
$$= -i\partial_\mu\bar{\psi}\gamma^\mu - m\bar{\psi} \tag{5}$$

(Derivative acts on the left)

- $k_\mu^2 \equiv k_\mu k^\mu$
- ψ^α left handed Weyl spinor
- $\tilde{\psi}_{\dot{\alpha}}$ right handed Weyl spinor
- $\psi = \begin{pmatrix} \psi^\alpha \\ \tilde{\chi}_{\dot{\beta}} \end{pmatrix}$ Dirac spinor (Representation of $\left(\frac{1}{2}, 0\right) \oplus \left(0, \frac{1}{2}\right)$)

All of these expressions regarding spinors will be explained or will quickly become evident in Section 2, due to the complexity of their structure.

1.2. Integrals that are useful

- $$\int \mathrm{d}x\ f(x)\delta(x^2-a^2) = \frac{1}{2|a|}(f(a)+f(-a)) \tag{6}$$
- $$\int \mathrm{d}^3k\ \theta(k^0)\delta(k_\mu^2-m^2) = \frac{1}{2\omega_k}[\theta(\omega_k)+\theta(-\omega_k)] = \frac{1}{2\omega_k} \tag{7}$$

  where $\omega_k = \sqrt{\vec{k}^2 - m^2}$
- $$\int \frac{\mathrm{d}^4k}{(2\pi)^4} \exp(ik(x-y)) = \delta^4(x-y) \tag{8}$$
- $$\int \frac{\mathrm{d}^Nk}{(2\pi)^N} \exp(ik(x-y)) = \delta^N(x-y) \tag{9}$$
- $$\int \mathrm{d}x \exp\left(-\frac{1}{2}ax^2 + Jx\right) = \left(\frac{2\pi}{a}\right)^{\frac{1}{2}} \exp\left(\frac{J^2}{2a}\right) \tag{10}$$

1.3. Group Theory

Definition 1.3.1: A Group G is a set of elements $\{g_i\} \in G$ and a rule $g_i \times g_j = g_k$ giving how they "multiply."

Group Theory Axioms

- For each element $g_i \in G$ and $g_j \in G$, their product $g_i \times g_j = g_k \in G$ is an element of G
- Products must be Associative:

$$(g_i \times g_j) \times g_k = g_i \times (g_j \times g_k) \tag{11}$$

- There must be an Identity:

$$\mathbb{1} \times g_i = g_i \times \mathbb{1} = g_i \tag{12}$$

- Elements have an Inverse:

$$g_i^{-1} \times g_i = \mathbb{1} \tag{13}$$

Definition 1.3.2 (Representation): An embedding of g_i as operators in a vector space

- Matrices are finite-dimensional representations of a group

Definition 1.3.3 (Faithful Representation): Each element in a group gets its own matrix

1.3.1. Lie Groups

- A Group G whose elements can be written as $g_i = \exp(ic_i^g \lambda_i)$ for some number c_i^g and some matrix λ_i (known as a *generator*)
- In an algebra, you can add and multiply
- In a Group, you have to follow the previously mentioned rules (just multiplication is allowed)

Definition 1.3.1.1 (Lie Group): A *Lie Group* is a type of group with an infinite number of elements and a finite number of generators.

Definition 1.3.1.2 (Lie Algebra): The Generators of a *Lie group* form a *Lie Algebra*.

Example 1.3.1.1 (Lie Algebras):
- QED : $\mathrm{U}(1)$ (unitarity)
- Weak: $\mathrm{SU}(2)$ (special unitarity)
- Strong: $\mathrm{SU}(3)$
- Lorent Group: $\mathrm{O}(1, 3)$ (orthogonal and preserving (1,3) metric) (see §2.2)

Part II

Scalar Field Theory

- Scalar Field Theory is *a* theory of quantum fields. It is mostly used as a toy theory to base other theories off of, since out of the many *many* particles that exist (dozens counting baryons), there is an entirety of **1** scalar particle (the Higgs boson). Therefore, Scalar Field Theory primarily is only useful to base other models of the universe off of it as opposed to modeling the universe after it.

2.1. Classical Field Theory

- The idea behind **classical field theory** is to utilize the idea of Lagrangians that will be presented in *Lagrangians* and in *Classical Symmetries* to obtain a "toy theory" without the quantum effects that will later appear such as Loops in diagrams, etc. In practice, this means that it is essentially the *Lagrangian Formalism* of Classical Mechanics where *position* and *momentum* coordinates (p, q) are replaced by fields (ϕ, π), as seen before in *Lagrangians* without having been explicitly stated.
- Classical Field Theory, much like *Classical Mechanics*, functions as the unquantized equivalent to Quantum Field Theory. Quantization does not require much to achieve as simply putting quantization conditions on the operators being used acheives it with relative ease.

Example 2.1.1 (Klein Gordon Equation):

- A very standard equation for a field in Classical Field theory is the *Klein Gordon Equation*. (This, in fact, describes a *free* field in which there are no interactions, allowing for a very good "toy theory" to use and potentially apply later).

$$(\Box + m^2)\phi = 0 \tag{14}$$

or

$$(\partial^2 + m^2)\phi = 0 \tag{15}$$

is the *Klein Gordon Equation*.

2.1.1. Lagrangians

Definition 2.1.1.1 (Lagrangian): The *Lagrangian* (Density) $\mathcal{L}$, similar to the *Hamiltonian* H (or $\mathcal{H}$ for Hamiltonian density, defined as $H = \int \mathrm{d}^3 x \mathcal{H}$) as they both can be used to describe the evolution of a system. In Field Theories, densities are used more commonly. The Density of either quantity, as stated before, is a more useful quantity in Field theories due to Lorentz Invariance and other such things.

Definition 2.1.1.2 (Action): Action S is defined as

$$S = \int \mathrm{d}t L = \int \mathrm{d}^4 x \mathcal{L} \tag{16}$$

which should be familiar from studying classical mechanics (Lagrangian mechanics specifically). This quantity is minimized in order to obtain the evolution of a system (**Equations of Motion**).

Theorem 2.1.1.1 (Euler Lagrange Equations): In order to minimize the action of a system with Lagrangian Density $\mathcal{L}[\phi, \partial_\mu \phi]$, the system must satisfy the ***Euler Lagrange Equations***:

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = 0 \tag{17}$$

A differential equation which ends up being extremely useful in physics for describing the time evolution of a system.

Proof: We aim to equate the variation of the action S with respect to the field ϕ to zero. That is, we require

$$\frac{\delta S}{\delta \phi} = 0 \tag{18}$$

Note that

$$S = \int \mathrm{d}t L = \int \mathrm{d}^4 x \mathcal{L}[\phi, \partial_\mu \phi] \tag{19}$$

in order to obtain δS, we must perform a transformation $\phi \to \phi + \delta\phi$ in the Lagrangian. This provides

$$\delta S = \int d^4 x \left(\frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta(\partial_\mu \phi) \right) \tag{20}$$

Via the product rule and the fact that infinitesimal variations commute with partial derivatives[1], we have that

[1]Look into variational calculus if you want to learn more about this

$$
\begin{aligned}
\delta S &= \int \mathrm{d}^4x \left(\frac{\partial \mathcal{L}}{\partial \phi}\delta\phi + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\delta\phi \right) - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\delta\phi \right) \\
\delta S &= \int \mathrm{d}^4x \left(\frac{\partial \mathcal{L}}{\partial \phi}\delta\phi - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\delta\phi + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\delta\phi \right) \right) \qquad (21) \\
\delta S &= \int \mathrm{d}^4x \left(\left(\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right)\delta\phi + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\delta\phi \right) \right)
\end{aligned}
$$

Now we need to prove that the final term (the one without the shared common factor of $\delta\phi$) vanishes as our integration bounds become infinite (as we are integrating over all spacetime of course). Since this term is a total derivative of some quantity that we will define to be

$$
A^\mu \equiv \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\delta\phi \qquad (22)
$$

(Something similar with the form of $\frac{A^\mu}{\delta\alpha}$ is a physically important quantity but this A^μ is not of any significance besides being useful for math). With the derivative being of course $\partial_\mu A^\mu$. The integral of this quantity is

$$
\int \mathrm{d}^4x \partial_\mu A^\mu \qquad (23)
$$

and by the **4 dimensional divergence theorem**, we obtain

$$
\int_V \mathrm{d}^4x \partial_\mu A^\mu = \oint_{\partial V} \mathrm{d}^3\sigma_\mu A^\mu \qquad (24)
$$

This must vanish, given that the $\delta\phi$ is forced to vanish at the boundaries by the principle of least action. Therefore, we have that

$$
\frac{\delta S}{\delta \phi} = \int \mathrm{d}^4x \left(\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) = 0 \qquad (25)
$$

For the integral to vanish, the integrand must vanish, therefore

$$
\boxed{\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = 0} \qquad (26)
$$

■

Theorem 2.1.1.2 (Extended Euler Lagrange Equations): In order to minimize the action of a system with Lagrangian Density $\mathcal{L}\left[\phi, \partial_\mu \phi, \partial_\nu \partial_\mu \phi, ...\right]$, the system must satisfy the ***Extended Euler Lagrange Equations***:

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} + \partial_\nu \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} - \cdots = 0 \tag{27}$$

(This was actually a problem in the Schwartz QFT textbook, so if you are reading that, potentially skip this part.)

Proof: We begin in a similar way to the proof to the *Euler-Lagrange equations.*

$$\delta S = \int \mathrm{d}^4 x \left(\frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta(\partial_\mu \phi) + \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} \delta(\partial_\nu \partial_\mu \phi) + \cdots \right) \tag{28}$$

We then utilize the product rule (or the reverse of it) to obtain

$$\begin{aligned}\partial_\nu \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} \delta \phi \right) = \partial_\nu \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} \delta \phi \\ + \partial_\nu \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} \delta(\partial_\mu \phi) \\ + \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} \delta(\partial_\nu \phi) \\ + \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} \delta(\partial_\nu \partial_\mu \phi)\end{aligned} \tag{29}$$

We can rewrite $\partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} \delta(\partial_\nu \phi)$ and $\partial_\nu \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} \delta(\partial_\mu \phi)$ via similar methods and we can continue to do so with more terms forming a pattern of the nth (with n partial derivatives) term has a coefficient of $(-1)^n$ (this method similar to the *principle of inclusion-exclusion*)

$$\begin{aligned}\delta S = \int \mathrm{d}^4 x \Bigg(\left(\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} + \partial_\nu \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} - \cdots \right) \delta \phi \\ + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta \phi \right) + \partial_\mu \left(\partial_\nu \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} \delta \phi \right) + \partial_\nu \left(\partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu)} \delta \phi \right) \\ + \partial_\mu \partial_\nu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \partial_\nu)} \delta \phi \right) + \cdots \Bigg)\end{aligned} \tag{30}$$

Every term besides those with the shared factor of $\delta \phi$ disappears due to similar reasons mentioned in the proof of the *Euler-Lagrange equations* (i.e. they are total derivatives). and therefore, since we want $\frac{\delta S}{\delta \phi} = 0$, we have that

$$\boxed{\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} + \partial_\nu \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} - \cdots = 0} \tag{31}$$

■

Example 2.1.1.1 (Lagrangian): An example of a Lagrangian that can apply to some general scalar field ϕ[2] is the Lagrangian

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial_\mu \phi) - V[\phi] \tag{32}$$

This would give us the equation of motion

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} &= 0 \\ -V'[\phi] - \Box \phi &= 0 \end{aligned} \tag{33}$$

so

$$(\Box + V'[\phi])\phi = 0 \tag{34}$$

which is related to the *Klein-Gordon equation* $(\Box + m^2)\phi = 0$.

- One can split the terms of a Lagrangian $\mathcal{L}$ into *Kinetic* terms and *Interacting* terms.
  - *Kinetic* terms of a Lagrangian are ones which have at most 2 fields in it
    - $$\mathcal{L}_{\text{kin}} \supset \frac{1}{2}\phi \Box \phi, \bar{\psi} \not{\partial} \psi, \frac{1}{4} F_{\mu\nu}^2, \frac{1}{2} m^2 \phi^2, \cdots \tag{35}$$
- Interacting terms are terms with more than 2 fields

$$\mathcal{L}_{\text{int}} \supset \lambda \phi^3, g\bar{\psi} \not{A} \psi, g\partial_\mu \phi A_\mu \phi^\star, \cdots \tag{36}$$

> **Note**: Interacting terms in the lagrangian all have a factor λ or g associated with them. This is due to the fact that it is a coupling constant related to the interaction.

2.1.2. Classical Symmetries

2.1.2.1. Noether's Theorem

- This theorem is the core in the utilizing symmetries in classical and non-classical field theory.

[2] That is, it doesn't apply universally, but just in this circumstance

Theorem 2.1.2.1.1 (Noether's Theorem): If a Lagrangian has a continuous symmetry (i.e. it is parameterizable by some continuous α), there exists a conserved current J^μ associated with said symmetry when the equations of motion of the Lagrangian are satisfied, i.e. the **Euler-Lagrange** equations. In other words, *symmetries imply conservation laws.*

Proof: The Proof for *Noether's Theorem* is fairly short and clever.

We define $\mathcal{L}$ as the Lagrangian of the system, ϕ_n as the field(s) involved in the system, and α as the parameter of the continuous symmetry.

Similar to the derivation of the *Euler-Lagrange* equations,

$$0 = \frac{\delta \mathcal{L}}{\delta \alpha} = \sum_n \left(\left(\frac{\partial \mathcal{L}}{\partial \phi_n} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \right) \frac{\delta \phi_n}{\delta \alpha} + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \frac{\delta \phi_n}{\delta \alpha} \right) \tag{37}$$

This expression must be equivalent to zero due to α parametrizing a symmetry (i.e. the Lagrangian is invariant wrt some variation in the parameter). It is necessary to define a

$$J^\mu \equiv \sum_n \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \frac{\delta \phi_n}{\delta \alpha} \tag{38}$$

Since we are only considering the case in which the equations of motion hold true, (i.e. the *Euler-Lagrange equations* are true), we have that

$$\boxed{\partial_\mu J^\mu = \partial_\mu \sum_n \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \frac{\delta \phi_n}{\delta \alpha} = \sum_n \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \frac{\delta \phi_n}{\delta \alpha} = 0} \tag{39}$$

Therefore due to some continuous symmetry parametrized by some α, there exists a conserved current J^μ. ∎

- The *Energy-Momentum Tensor* $\mathcal{T}_{\mu\nu}$ is the conserved current for spacetime translation.

Definition 2.1.2.1.1 (Energy Momentum Tensor): The *Energy-Momentum Tensor* $\mathcal{T}_{\mu\nu}$ for some Lagrangian $\mathcal{L}[\phi_1, \partial_\mu \phi_1, \dots \phi_n, \partial_\mu \phi_n]$ and some spacetime translation $x^\nu \to x^\nu + \xi^\nu$ is defined as

$$\mathcal{T}_{\mu\nu} \equiv \sum_n \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \partial_\nu \phi_n - g_{\mu\nu} \mathcal{L} \tag{40}$$

Derivation below.

Starting with the translation

$$\phi(x) \to \phi(x + \xi) = \phi(x) + \partial_\nu \xi^\nu \phi(x) + \dots \tag{41}$$

This provides $\frac{\delta\phi}{\delta\xi^\nu} = \partial_\nu \phi$ which applies universally regardless of the field ϕ (and in fact regardless of the physical object, as it applies to the Lagrangian $\mathcal{L}$, a consequence of the fact that $\delta S = 0$).

We then have that

$$\begin{aligned} \frac{\delta\mathcal{L}}{\delta\xi^\nu} &= \partial_\mu \left(\sum_n \frac{\partial\mathcal{L}}{\partial(\partial_\mu \phi_n)} \frac{\delta\phi_n}{\delta\xi^\nu} \right) \\ &= \partial_\mu \left(\sum_n \frac{\partial\mathcal{L}}{\partial(\partial_\mu \phi_n)} \partial_\nu \phi_n \right) \\ \partial_\nu \mathcal{L} &= \partial_\mu \left(\sum_n \frac{\partial\mathcal{L}}{\partial(\partial_\mu \phi_n)} \partial_\nu \phi_n \right) \end{aligned} \tag{42}$$

We then have that

$$\partial_\mu \left(\sum_n \frac{\partial\mathcal{L}}{\partial(\partial_\mu \phi_n)} \partial_\nu \phi_n - g_{\mu\nu}\mathcal{L} \right) = 0 \tag{43}$$

Where we then obtain

$$\mathcal{T}_{\mu\nu} \equiv \left(\sum_n \frac{\partial\mathcal{L}}{\partial(\partial_\mu \phi_n)} \partial_\nu \phi_n - g_{\mu\nu}\mathcal{L} \right) \tag{44}$$

and $\partial_\mu \mathcal{T}_{\mu\nu} = 0$.

- Energy density:

$$\mathcal{E} = \mathcal{T}_{00} = \sum_n \frac{\partial\mathcal{L}}{\partial(\dot{\phi}_n)} \dot{\phi} - \mathcal{L} \tag{45}$$

  - $\dot{\phi}_n = \partial_t \phi_n$
  - In the case of Lagrangians with $\dot{\phi} = \pi = \frac{\partial\mathcal{L}}{\partial\dot{\phi}}$, $\mathcal{E} = \mathcal{H}$ due to the *Legendre Transform*

Definition 2.1.2.1.2 (4-Momentum): In terms of the energy-momentum tensor, 4-momentum p^μ is defined as

$$p^\nu \equiv \int \mathrm{d}^3 x \mathcal{T}^{0\nu} = (E, \vec{p}) \tag{46}$$

2.1.2.2. More information on currents

- A current can refer to a Noether current (such as the ones previously seen)
- They can be external, such as a moving electron in a wire
- They can be sources for fields
  - They are never fields with kinetic terms
- They can be placeholders in Lagrangians

Example 2.1.2.2.1 (Currents in Lagrangians): In

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \tilde{\phi}\Box\phi - ieA_\mu\left(\tilde{\phi}\overleftrightarrow{\partial}_\mu\phi\right), \tag{47}$$

we can write it as

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \tilde{\phi}\Box\phi - A_\mu J^\mu \tag{48}$$

- Clearly, you can see how currents are used to represent quantities in lagrangians that we do not care for.

2.2. Scalar Field Theory

- Scalar Field Theory is fairly similar to Classical Field Theory with the exception that it is quantized via the methods that will be presented shortly. This ensures that it obeys the laws of quantum mechanics, while still being a "Toy Theory" due to the simplicity of the objects involved in it ((potentially) complex fields ϕ as opposed to *vectors* A_μ or *spinors* ψ which are closer to real, physical particles).

2.2.1. Quantization

- As seen in ‘s Quantum Field Theory Notes, one could try to naively try to quantize the Schrodinger Equation (via the use of a series to have the equation be on-shell), which would bring rise to some problems

$$\begin{gathered} H|\psi\rangle = i\partial_t|\psi\rangle \\ H = \sqrt{\hat{p}^2 + m^2} \end{gathered} \tag{49}$$

2.2.1.1. Harmonic Oscillator/First Quantization

- The standard Hamiltonian for a simple harmonic oscillator is

$$H = \frac{1}{2}\frac{p^2}{2m} + \frac{1}{2}m\omega^2x^2 \tag{50}$$

. The relevance to quantization will become apparent.

Definition 2.2.1.1.1 (First Quantization): The idea behind first quantization is to apply the commutation relation $[\hat{x}, \hat{p}] = i$ to a system. This commutation relation is known as the *canonical commutation relation*, and it can be used to obtain quantum mechanics from a classical theory (which we are doing now).

- Utilizing a change of variables to a and $a^\dagger$, one can derive an alternative form for the Hamiltonian

$$a \equiv \sqrt{\frac{m\omega}{2}}\left(x + \frac{ip}{m\omega}\right),\ a^\dagger \equiv \sqrt{\frac{m\omega}{2}}\left(x - \frac{ip}{m\omega}\right) \tag{51}$$

- $$H = \omega\left(a^\dagger a + \frac{1}{2}\right) \tag{52}$$

  is written in a very suggestive form. Such $a^\dagger$ and a can be thought of as "exciting" the harmonic oscillator, as you may know from quantum mechanics.
- The eigenstates of this Hamiltonian are eigenstates of $\hat{N} \equiv a^\dagger a$

- Notice that this hamiltonian's eigenvalues (the energies of the eigenstates) are discrete. This is a result from quantum mechanics (or, rather, defining quantum mechanics). Such a result is the reason behind both the name *Quantum Mechanics* and the idea of *First Quantization.*
- $\hat{N}$ operator:
  - $\hat{N}|n\rangle = n|n\rangle$
  - $a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$
  - $a|n\rangle = \sqrt{n}|n-1\rangle$

2.2.1.2. Heisenberg Picture

- $$i\frac{\mathrm{d}}{\mathrm{d}t}a = [a, H] = \left[a, a\left(a^\dagger a + \frac{1}{2}\right)\right] = \omega\left(aa^\dagger a - a^\dagger aa\right) = \omega\left[a, a^\dagger\right]a = \omega a \tag{53}$$
- Solution is

$$a(t) = e^{-i\omega t}a(0) \tag{54}$$

2.2.1.3. Relativistic Fields

- A common relativistic equation (heavy foreshadowing) for a (free) field is the equation

$$\square\phi = 0. \tag{55}$$

  - Solution to this is

$$\phi(x,t) = \int \frac{\mathrm{d}^3p}{(2\pi)^3}\left[a_p(t)e^{i\vec{p}\cdot\vec{x}} + \tilde{a_p}(t)e^{-i\vec{p}\cdot\vec{x}}\right] \tag{56}$$

  which is written in a fairly suggestive manner

2.2.1.4. 2nd Quantization

> **Definition 2.2.1.4.1** (Second Quantization): To obtain 2nd quantization (and hence quantum field theory), one can use an annihilation operator a_p and a creation operator $a_p^\dagger$ which exist per wave number as opposed to being universal since now, we utilize infinite harmonic oscillators which exist per p

- Free Hamiltonian is obtained via integration over these

$$H = \int \frac{\mathrm{d}^3p}{(2\pi)^3}\omega_p\left(a_p^\dagger a_p + \frac{1}{2}\right) \tag{57}$$

with $\omega_p = |\vec{p}|$

- 2nd Quantization new things:
  - Each (of the infinitely many) point of momentum $\vec{p}$ gets its own harmonic oscillator (which we integrate over to obtain our result)
  - The nth excitation of the harmonic oscillator $\vec{p}$ as being a state with n particles

2.2.1.5. Fock Space

- Instead of Hilbert space $\mathcal{H}$, like in Quantum Mechanics/First Quantization, we have a *Fock Space* $\mathcal{F}$.
- A Fock space $\mathcal{F}$ is the direct sum of Hilbert spaces $\mathcal{H}_n$, that is,

$$\mathcal{F} = \bigoplus_n \mathcal{H}_n \tag{58}$$

- States in $\mathcal{H}_n$ are linear combinations of $\left\{ \left| p_1^\mu ... p_n^\mu \right\rangle \right\}$
  - Since we are using *Fock Space*, this means that there can be however many particles due to the direct sum over Hilbert Spaces

Note (Momentum Notation): $p^0 = \sqrt{\vec{p}^2 + m^2}$ so $|p\rangle = |\vec{p}\rangle = |p^\mu\rangle$ due to no loss of generality (this only works if it is on-shell).

2.2.1.6. Field Expansion

- Previously, we noted that $\left[a, a^\dagger\right] = 1$. This generalizes to $\left[a_k, a_p^\dagger\right] = (2\pi)^3 \delta^3\left(\vec{p} - \vec{k}\right)$
- $a_p^\dagger |0\rangle = \frac{1}{\sqrt{2\omega_p}} |p\rangle$
- $\langle 0|0\rangle = 1$ since it is the vacuum ground state
  - $\langle p|k\rangle = 2\sqrt{\omega_p \omega_k} \left\langle 0 \middle| a_p a_o^\dagger \middle| 0 \right\rangle = 2\omega_p (2\pi)^3 \delta^3\left(\vec{p} - \vec{k}\right)$
- Identity operator

$$\mathbb{1} = \int \frac{\mathrm{d}^3 p}{(2\pi)^3} \frac{1}{2\omega_p} |p\rangle \langle p| \tag{59}$$

- Free field definition

$$\phi_0(\vec{x}) \equiv \int \frac{\mathrm{d}^3 p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \left(a_p e^{i\vec{p}\cdot\vec{x}} + a_p^\dagger e^{-i\vec{p}\cdot\vec{x}}\right) \tag{60}$$

which is why I said $\phi(\vec{x}, t)$ from 4.1.2 was written in a suggestive manner (although this is still technically just a definition)

Note (Momentum state inner product with free field):

$$\begin{aligned}\langle\vec{p}|\phi_0(\vec{x})|0\rangle &= \langle 0|\sqrt{2\omega_p}a_p \int \frac{\mathrm{d}^3k}{(2\pi)^3}\left(a_k e^{i\vec{k}\vec{x}} + a_k^\dagger e^{-i\vec{k}\cdot\vec{x}}\right)|0\rangle \\ &= \int \frac{\mathrm{d}^3k}{(2\pi)^3}\sqrt{\frac{\omega_p}{\omega_k}}\left[e^{i\vec{k}\cdot\vec{x}}\langle 0|a_p a_k|0\rangle + e^{-i\vec{k}\cdot\vec{x}}\left\langle 0\middle|a_p a_k^\dagger\middle|0\right\rangle = e^{-i\vec{p}\cdot\vec{x}}\right] \\ &= e^{-i\vec{p}\cdot\vec{x}}\end{aligned} \tag{61}$$

This is the expected result of $\langle\vec{p}|\vec{x}\rangle$, and so we conclude $\phi_0(\vec{x})|0\rangle = |\vec{x}\rangle$

2.2.1.7. Time Dependence

- Because of Heisenberg's equations of motion, in the Heisenberg perspective, $a_p(t)$ and $a_p^\dagger(t)$ are time dependent
  - $a_p(t) = e^{-i\omega_p t}a_p, a_p^\dagger(t) = e^{i\omega_p t}a_p^\dagger$
- Heisenberg equations of motion
  - $$\begin{aligned}[H_0, \phi_0(\vec{x}, t)] &= \int \frac{\mathrm{d}^3p}{(2\pi)^3}\int \frac{\mathrm{d}^3k}{(2\pi)^3}\frac{1}{\sqrt{2\omega_k}}\left[\omega_p\left(a^\dagger a_p + \frac{1}{2}\right), a_k e^{-ikx} + a_k^\dagger e^{ikx}\right] \\ &= \int \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{1}{\sqrt{2\omega_p}}\left[-\omega_p a_p e^{-ipx} + \omega_p a_p^\dagger e^{ipx}\right] \\ &= -i\partial_t\phi_0(\vec{x}, t)\end{aligned} \tag{62}$$
  - Should have that, for any field ϕ,

$$[H, \phi(\vec{x}, t)] = i\partial_t\phi(\vec{x}, t) \tag{63}$$

  - $$\phi(\vec{x}, t) = \int \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{1}{\sqrt{2\omega_p}}\left[a_p(t)e^{-ipx} + a_p^\dagger(t)e^{ipx}\right] \tag{64}$$

    is the solution to the *Heisenberg equations of motion* for ϕ, our field which does not necessarily have to be free in this case

2.3. Cross Sections

- Cross sections are how physicists analyze collider data (such as the exabytes of LHC data that is collected every year)

Important Result of this Section

$$\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{\mathrm{CM}} = \frac{1}{64\pi^2 E_{\mathrm{CM}}^2}|\mathcal{M}|^2 \tag{65}$$

and the Born approximation

$$\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{\mathrm{Born}} = \frac{m_e^2}{4\pi^2}\left[\tilde{V}\left(\vec{k}\right)\right]^2 \tag{66}$$

- Cross sections used to make sense of particle accelerator data

- Cross sectional area:

$$\begin{aligned}\sigma &= \frac{\#\text{ particles}}{\text{time} \times \text{number density} \times \text{velocity of beam}} \\ &= \frac{1}{T}\frac{1}{\Phi}N, \Phi = \#\text{ density} \times \text{beam vel.} = \frac{1}{V} \times |\vec{v}|\end{aligned} \tag{67}$$

  - More useful/natural is differential cross section

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} \tag{68}$$

    for some angle/change in angle Ω.
  - Info about shape of object with this

2.3.0.1. QM

- Generalize to Differential Cross Sections due to their changing based on angle in colliders
- In QM, there is a probability of scattering

$$P = \frac{N}{N_{\text{inc}}} \tag{69}$$

  with N_{inc} number of incident particles
- Cross section

$$d\sigma = \frac{1}{T}\frac{1}{\Phi}dP \tag{70}$$

  (normalize flux)
- Differential scattering number

$$\mathrm{d}N = L \times \mathrm{d}\sigma \tag{71}$$

  - L is defined as the *Luminosity* through this equation

> **Note** (Differential solid angle): The cross section $\mathrm{d}\sigma$ is generally measured via $\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}$ as seen earlier. This $\mathrm{d}\Omega$ is equivalent to
>
> $$\mathrm{d}\Omega = \sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi = -\,\mathrm{d}\cos\theta\,\mathrm{d}\phi \tag{72}$$

2.3.1. Cross Section to Matrix Element

- $2 \to N$ Particle Interaction:

$$p_1 + p_2 \to \{p_j\} \tag{73}$$

- $$\begin{aligned}\Phi &= \frac{|\vec{v}_1 - \vec{v}_2|}{V} \\ \Longrightarrow \mathrm{d}\sigma &= \frac{V}{T}\frac{1}{|\vec{v}_1 - \vec{v}_2|}\,\mathrm{d}P\end{aligned} \tag{74}$$

- Have that

$$\mathrm{d}P = \frac{|\langle f|S|i\rangle|^2}{\langle f|f\rangle\langle i|i\rangle}\,\mathrm{d}\Pi \tag{75}$$

- with

$$\mathrm{d}\Pi = \prod_j \frac{N}{(2\pi)^3}\mathrm{d}^3 p_j \tag{76}$$

- $$\int \mathrm{d}\Pi = 1 \tag{77}$$

  since

$$\int \frac{\mathrm{d}P}{(2\pi)} = \frac{1}{V} \tag{78}$$

- $$\left[a_p, a_q^\dagger\right] = (2\pi)^3\,\delta^3(p-q) \tag{79}$$
  - Implies that $\langle p|p\rangle = (2\pi)^3\,\delta^3(0)$
  - Normally $\delta^3(0)$ is infinite but

$$\begin{aligned}(2\pi)^3\,\delta^3(p) &= \int_V \mathrm{d}^3 x e^{i\vec{p}\cdot\vec{x}} \\ \implies \delta^3(0) &= \frac{V}{(2\pi)^3} \\ \text{and } \delta^4(0) &= \frac{TV}{(2\pi)^4}\end{aligned} \tag{80}$$

- $$\langle p|p\rangle = 2\omega_p V = 2E_p V \tag{81}$$
- Since

$$|i\rangle = |p_1\rangle|p_2\rangle \tag{82}$$

  and

$$|f\rangle = \prod_j |p_j\rangle \tag{83}$$

  - It is implied that

$$\langle i|i\rangle = (2E_1 V)(2E_2 V) \tag{84}$$

  - It is also implied that

$$\langle f|f\rangle = \prod_j 2E_j V \tag{85}$$

2.3.2. Actually Calculating the S-matrix (perturbatively)

- Notice that the *first order* of a S-matrix is just $\mathbb{1}$ since that is the case in which nothing happens
- Write $S = \mathbb{1} + i\mathcal{T}$
  - $\mathcal{T} \equiv$ **transfer matrix**

- $\mathcal{T}$ implies momentum conservation since it vanishes unless $|i\rangle$ and `ket` $* f\rangle$ have the same 4-momentum
- $\mathcal{T} = (2\pi)^4\,\delta^4(\Sigma p)\mathcal{M}$

2.3.2.1. Matrix Elements

- $\langle f|S - \mathbb{1}|i\rangle = i(2\pi)^4\,\delta^4(\Sigma P)\langle f|\mathcal{M}|i\rangle$
  - $$\begin{aligned} |\langle f|S|i\rangle|^2 &= \delta^4(0)\delta^4(\Sigma p)(2\pi)^8|\langle f|M|i\rangle|^2 \\ &= \delta^4(\Sigma p)TV(2\pi)^4|\mathcal{M}|^2 \end{aligned} \tag{86}$$

2.3.3. Lorentz Invariant Phase Space

- We denote the differential $\mathrm{d}\Pi_{\text{LIPS}}$ as being defined as the product over final momenta differentials times delta function factors, that is

$$d\Pi_{\text{LIPS}} \equiv \prod_{\text{final } j} \frac{\mathrm{d}^3 p_j}{(2\pi)^3} \frac{1}{2E_{p_j}} (2\pi)^4\,\delta^4(\Sigma p) \tag{87}$$

- Utilizing this, (as well as the derived S-matrix elements from earlier) we can write the differential probability $\mathrm{d}p$ via

$$\mathrm{d}P = \frac{|\langle f|S|i\rangle|^2}{\langle f|f\rangle\langle i|i\rangle}\,\mathrm{d}\Pi = \frac{(2\pi)^4\,\delta^4(\Sigma p) \times TV}{2E_1V \times 2E_2V} \times \frac{1}{\left(\Pi_j\left(2E_jV\right)|\mathscr{M}|^2 \prod_j \frac{\mathrm{d}^3 p_j}{(2\pi)^3} V = T\frac{1}{2E_1 \times 2E_2}|\mathscr{M}|^2\,\mathrm{d}\Pi_{\text{LIPS}}\right.} \tag{88}$$

- This easily gives us the cross section for $2 \to j$ decay:

$$\mathrm{d}\sigma = \frac{1}{((2E_1)(2E_2)|\vec{v}_1 - \vec{v}_2||\mathscr{M}|^2\,\mathrm{d}\Pi_{\text{LIPS}}} \tag{89}$$

2.3.4. Particle Decay Rate

- This is another useful quantity in analyzing accelerator data denoted by $\mathrm{d}\Gamma$
- The definition for this decay rate is $\mathrm{d}\Gamma \equiv \frac{1}{T}\,\mathrm{d}P$ (probability of decay in certain amount of time)
  - Specifically it is the probability that a particle of momentum p_1 decays into particles of momenta p_j in a time T
  - S-matrix theory cannot be used to describe these decays (?)
- $\mathrm{d}\Gamma = \frac{1}{2E_1}|\mathscr{M}|^2\,\mathrm{d}\Pi_{\text{LIPS}}$

2.3.5. Cross Section formula

- Consider $2 \to 2$ Scattering
  - This means that the momenta are $p_1 + p_2 \to p_3 + p_4$
- In the center of mass frame, the net momentum for each is zero. Therefore $\boldsymbol{p}_1 = -\boldsymbol{p}_2, \boldsymbol{p}_3 = -\boldsymbol{p}_4$ which implies that $E_1 + E_2 = E_3 + E_4 \equiv E_{\text{CM}}$
- Then, for the LIPS differential, we get that

$$\mathrm{d}\Pi_{\text{LIPS}} = (2\pi)^4\,\delta^4(\Sigma p)\frac{\mathrm{d}^3 p_3}{(2\pi)^3}\frac{1}{2E_3}\frac{\mathrm{d}^3 p_4}{(2\pi)^3}\frac{1}{2E_4} \tag{90}$$

- This is equivalent to the integral

$$\mathrm{d}\Pi_{\mathrm{LIPS}} = \frac{1}{16\pi^2}\,\mathrm{d}\Omega \int \mathrm{d}p_f \frac{p_f^2}{E_3}\frac{1}{E_4}\delta(E_3 + E_4 - E_{\mathrm{CM}}) \tag{91}$$

(I don't understand how this is true)
  - Where $p_f = |\boldsymbol{p_3}| = |\boldsymbol{p_4}|$ and $E_3 = \sqrt{m_3^2 + p_f^2}$, $E_4 = \sqrt{m_4^2 + p_f^2}$
  - Notice that the energies are dependent on the final momentum p_f, and so this integral is in fact, not trivial.
- Now we conduct a change of variables from p_f to $x(p_f) \equiv E_3(p_f) + E_4(p_f) - E_{\mathrm{CM}}$
  - Calculating the change of variables, we get

$$\frac{\mathrm{d}x}{\mathrm{d}p_f} = \frac{\mathrm{d}}{\mathrm{d}p_f}(E_3 + E_4 - E_{\mathrm{CM}}) = \frac{p_f}{E_3} + \frac{p_f}{E_4} = \frac{E_3 + E_4}{E_3 E_4} p_f \tag{92}$$

- We also have that

$$E_3 + E_4 = E_{\mathrm{CM}} \tag{93}$$

- Then,

$$\begin{aligned} \mathrm{d}\Pi_{\mathrm{LIPS}} &= \frac{1}{16\pi^2}\,\mathrm{d}\Omega \int_{m_3+m_4-E_{\mathrm{CM}}}^{\infty} \mathrm{d}x \frac{p_f}{E_{\mathrm{CM}}}\delta(x) \\ &= \frac{1}{16\pi^2}\,\mathrm{d}\Omega \frac{p_f}{E_{\mathrm{CM}}}\theta(E_{\mathrm{CM}} - m_3 - m_4) \end{aligned} \tag{94}$$

where $\theta(x)$ is the Heaviside step function due to the integral of the delta function.

- Putting this all together, we obtain that

$$\mathrm{d}\sigma = \frac{1}{(2E_1)(2E_2)|\boldsymbol{v}_1 - \boldsymbol{v}_2|}\frac{1}{16\pi^2}\,\mathrm{d}\Omega \frac{p_f}{E_{\mathrm{CM}}}|\mathscr{M}|^2\theta(E_{\mathrm{CM}} - m_3 - m_4) \tag{95}$$

  - Using

$$|\boldsymbol{v}_1 - \boldsymbol{v}_2| = \left|\frac{|\boldsymbol{p}_1|}{E_1} + \frac{|\boldsymbol{p}_2|}{E_2}\right| = |\boldsymbol{p}_i|\frac{E_{\mathrm{CM}}}{E_1 E_2} \tag{96}$$

  , we obtain that, under the condition that the masses are equivalent,

$$\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{\mathrm{CM}} = \frac{1}{64\pi^2 E_{\mathrm{CM}}^2}|\mathscr{M}|^2 \tag{97}$$

  as desired, allowing for computation of cross sections for simple scattering based off of quantum field theory predictions for $|\mathscr{M}|^2$

2.3.6. Born Approximation

- The Born approximation is a special case where a scalar electron ϕ_e with mass m_e scatters off of a scalar proton ϕ_p with mass m_p
  - It is considered the nonrelativistic limit
- I will not derive the born approximation, however, it is given by the formula

$$\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{\text{Born}} \equiv \frac{m_e^2}{4\pi^2}\left[\tilde{V}\left(\vec{k}\right)\right]^2 \quad (98)$$

where $\tilde{V}$ is the fourier transformed potential/the potential in fourier space.

▸
$$\tilde{V}\left(\vec{K}\right) = \int \mathrm{d}^3 x e^{-i\vec{k}\cdot\vec{x}} V(\vec{x}) \quad (99)$$

Note (Dimension of Matrix Elements):

- Utilizing the formula

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{1}{64\pi^2 E_{\text{CM}}^2}|\mathscr{M}|^2 \quad (100)$$

  , we can fairly clearly see that since $\left[\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right] = -2$ and that $\left[E_{\text{CM}}^2\right] = -2$, $\mathscr{M}$ must be dimensionless which should logically make sense.

2.4. Propagators

2.4.1. Green's Functions

Definition 2.4.1.1 (Green's Functions): A Green's function is a function of the form A^{-1} where A is an operator. A standard way to evaluate these green function operators is via a Fourier transform into k-space[3]. This is a type of *propagator*.

Definition 2.4.1.2 (Propagator): Defines or is used to define how a field propagates via applying the propagator Π to some other field or current. Can be used perturbatively in order to describe more complicated fields.

- An example of a propagator is the 2-point Green's function $\Pi = -\frac{1}{\Box}$ which is defined by

$$\Box_x \Pi(x, y) = -\delta^4(x - y) \quad (101)$$

- With $\Box_x = g^{\mu\nu}\frac{\partial}{\partial x^\mu}\frac{\partial}{\partial x^\nu}$ This gives the solution (via the Fourier transform), that

$$\Pi(x, y) = \int \frac{\mathrm{d}^4 k}{(2\pi)^4} e^{ik(x-y)}\frac{1}{k^2} \quad (102)$$

.

Note (Fourier Transform of Operators): Under *Fourier Transforms*, certain operators are effectively interchangeable with some counterpart that often simplifies calculations. An example of this is how $\Pi(x, y)$ is interchangeable with $\frac{1}{k^2}$.

[3] k-space is effectively momentum space due to the relation $p = \hbar k$ and the fact that we are using natural units

Example 2.4.1.1 (Delta Functions): For an instance of this occurrence, we can utilize delta functions.

Note: The Fourier transform of any delta function is

$$\tilde{\delta}\left(\vec{k}\right) = 1$$
$$\Longrightarrow \delta(\vec{x}) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} e^{i\vec{k}\cdot\vec{x}} \tag{103}$$

- $$\begin{aligned} \triangle^n \, \delta^3(\vec{x}) &= \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \, \triangle^n \, e^{i\vec{k}\cdot\vec{x}} \\ &= \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \left(-\vec{k}^2\right)^n e^{i\vec{k}\cdot\vec{x}} \end{aligned} \tag{104}$$
- $\triangle^n \longleftrightarrow \left(-\vec{k}^2\right)^n$ (this means that in a *Fourier Transform*, these are interchangeable).
- Additionally, we can do the same for the relativistic $\square = \partial_\mu \partial^\mu$

$$\square^n\left(\delta^4(x)\right) = \int \frac{\mathrm{d}^4 k}{(2\pi)^4} \square^n e^{ik\cdot x} = \int \frac{\mathrm{d}^4 k}{(2\pi)^4} \left(-k^2\right)^n e^{ik\cdot x} \tag{105}$$

- Have that $[\widetilde{\square^n \delta^4}](k) = \left(-k^2\right)^n$ and $[\widetilde{\triangle^n \, \delta^3}](k) = \left(-\vec{k}^2\right)^n$

2.4.2. Perturbation Theory

Definition 2.4.2.1 (Perturbation Theory): This topic is introduced in Quantum mechanics (and often before-hand) where we consider an initial state that we can solve and then higher order terms that we continue to be able to solve. Perturbation theory only works when the leading term is the "dominant" force in play. Often, we can utilize an initial term which is *non-interacting* and extra terms that are *interacting*.

Example 2.4.2.1 (Gravity): This example was provided from the Schwartz textbook[4] We start with the Lagrangian

$$\mathcal{L} = -\frac{1}{2} h \Box h + \frac{1}{3} \lambda h^3 + J h \tag{106}$$

where J is some *current* and h is our field. From the *Euler-Lagrange equations*, our equation of motion is

$$\Box h - \lambda h^2 - J = 0 \tag{107}$$

First order approx ($\lambda = 0$): $h_0 = \frac{1}{\Box} J$. We let $h = h_0 + h_1$.

$$\Box(h_0 + h_1) - \lambda (h_0 + h_1)^2 - J = 0 \tag{108}$$

implies

$$\begin{gathered} \Box h_1 = \lambda h_0^2 + \mathcal{O}(\lambda^2) \\ h = \frac{1}{\Box} J + \lambda \frac{1}{\Box} \left(\left(\frac{1}{\Box} J \right) \left(\frac{1}{\Box} J \right) \right) + \mathcal{O}(\lambda^2) \end{gathered} \tag{109}$$

which we now have in terms of our Green's Functions.

Note: In fact, something being defined as $\frac{1}{\Box} J$ is not that uncommon, as it also happens with the Electromagnetic field.

2.4.3. OFPT

- For **Old Fashioned Perturbation Theory** we utilize perturbation by setting the Hamiltonian to

$$H = H_0 + V \tag{110}$$

- The idea is that we have some $|\phi\rangle$ and some $|\psi\rangle$ such that

$$H_0 |\phi\rangle = E |\phi\rangle \tag{111}$$

and

$$H |\psi\rangle = E |\psi\rangle \tag{112}$$

- From this we can derive

$$|\psi\rangle = |\phi\rangle + \frac{1}{E - H_0} V |\phi\rangle, \tag{113}$$

the *Lippmann-Schwinger* equation
  - This allows us to define the propagator[5]

[4]This is the primary textbook that I am learning Quantum Field Theory through, as seen in the credits section.

[5]In fact, this is not well-defined, but the limit

$$\Pi_{\text{LS}} \equiv \frac{1}{E - H_0} \tag{115}$$

- We define an operator T as

$$\begin{aligned} V|\psi\rangle &= T|\phi\rangle \\ |\psi\rangle &= |\phi\rangle + \Pi_{\text{LS}} T|\phi\rangle \\ V|\psi\rangle &= V|\phi\rangle + V\Pi_{\text{LS}} T|\phi\rangle \end{aligned} \tag{116}$$

  - Note that $V|\psi\rangle = T|\phi\rangle$ so

$$T|\phi\rangle = V|\phi\rangle + V\Pi_{\text{LS}} T|\phi\rangle = (V + V\Pi_{\text{LS}} T)|\phi\rangle. \tag{117}$$

  (This holds when contracted with any eigenstate of H_0, $|\phi_j\rangle$)

- This implies that that $T = V + V\Pi_{\text{LS}} T$.
- Recursion provides

$$T = V + V\Pi_{\text{LS}} V + V\Pi_{\text{LS}} V\Pi_{\text{LS}} V + \cdots \tag{118}$$

If we want a transition amplitude for this, we want to calculate

$$\langle f|T|i\rangle = \langle f|V|i\rangle + \langle f|V\Pi_{\text{LS}} V|i\rangle + \cdots \tag{119}$$

> **Note**: We can insert the eigenstates within this equation due to the fact that
>
> $$\sum_j |\phi_j\rangle\langle\phi_j| = \mathbb{1} \tag{120}$$

This provides

$$\langle\phi_f|T|\phi_i\rangle = \langle\phi_f|V|\phi_i\rangle + \langle\phi_f|V\Pi_{\text{LS}}|\phi_j\rangle\langle\phi_j|V|\phi_i\rangle + \cdots \tag{121}$$

- In an alternative/more convenient notation,

$$T_{fi} = V_{fi} + \left(\sum_j\right) V_{fj}\Pi_{\text{LS}}(j)V_{ji} + \left(\sum_j\right) V_{fi}\Pi_{\text{LS}}(j)V_{jk}\Pi_{\text{LS}}(k)V_{ki} + \cdots \tag{122}$$

  Where $T_{ij} = \langle\phi_i|T|\phi_j\rangle$, $V_{ij} = \langle\phi_i|V|\phi_j\rangle$, $\Pi_{\text{LS}}(j) = \frac{1}{E-E_j}$ with $E_f = E_i$ and the sums tend to be omitted

> **Note** (OFPT With particle interactions): We can with a certain particle state $|i\rangle$ and end with a final certain particle state $|f\rangle$ and use *OFPT* to calculate what happens in-between such states. An example of this is the case of an electron scattering off an electron.

$$\lim_{\varepsilon\to 0} \frac{1}{E - H_0 + i\varepsilon} \tag{114}$$

is.

Example 2.4.3.1 (Electron Scattering): This is an example from the textbook that I was reading which allows for the derivation of the *Advanced* and *Retarded* propagators.

- The transition matrix element is

$$T_{fi} = V_{fi} + \sum_n V_{fn} \frac{1}{E_i - E_n} V_{ni} + \cdots \tag{123}$$

- For electron scattering, the initial and final states are $|i\rangle = |\psi_e^1 \psi_e^2\rangle$ and $\langle f| = \langle \psi_e^3 \psi_e^4|$
- Due to the fact that Coulomb's law is replaced by photon exchange we need a photon field $\phi(x)$.
- This provides

$$V = \frac{1}{2} e \int d^3 x \psi_e(x) \phi(x) \psi_e(x) \tag{124}$$

with the $\frac{1}{2}$ from spin and the e as some coupling constant (which happens to be the electron charge).

Note:
- This interaction turns a $\vec{p}_1$ electron into a $\vec{p}_3$ electron and a $\vec{p}_\gamma$ photon.
- Since both $|i\rangle$ and $|f\rangle$ have 2 electrons and no photons, $V_{ij} = 0$

2.4.3.1. Retarded and Advanced Propagators

- This gets split into two states. (propagated by the *Retarded* Propagator and the *Advanced* Propagator)

- The propagators that this creates are foundational to quantum field theory, (the advanced and retarded ones), which is the primary reason behind this methodology.

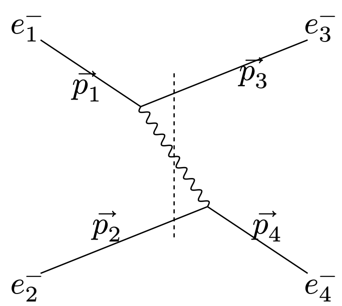

Figure 1: Retarded state

- In the Retarded state, electrons interact via the *Retarded Propagator*.

- We have that

$$|n\rangle = |\psi_e^3 \phi^\gamma \psi_e^2\rangle \tag{125}$$

  - Which implies that

$$
\begin{aligned}
V_{ni}^{R} &= \langle \psi_e^3 \phi^\gamma \psi_e^2 | V | \psi_e^1 \psi_e^2 \rangle \\
&= \langle \psi_e^3 \varphi^\gamma | V | \psi_e^1 \rangle \langle \psi_e^2 | \psi_e^2 \rangle \\
&= \langle \psi_e^3 \phi^\gamma | V | \psi_e^1 \rangle
\end{aligned} \tag{126}
$$

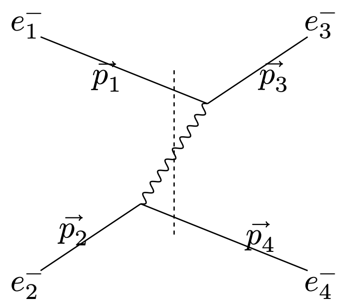

Figure 2: Advanced State

- For the *advanced state*, a similar result is derived
  - $$V_{ni}^{A} = \langle \psi_e^4 \varphi^\gamma | V | \psi_e^2 \rangle \tag{127}$$

2.4.3.2. Evaluating The Propagators

- $$V_{ni}^{R} = \langle \psi_e^3 \varphi^\gamma | V | \psi_e^1 \rangle = \frac{e}{2} \int \mathrm{d}^3 x \langle \psi_e^3 \varphi^\gamma | \psi_e(x) \phi(x) \psi_e(x) | \psi_e^1 \rangle \tag{128}$$

Note:
- Free field (like the photon and electron fields inside of the integral)

$$\langle \phi^\gamma | \phi(x) | 0 \rangle = e^{-i \vec{p} \cdot \vec{x}} \tag{129}$$

- This implies:

$$
\begin{aligned}
V_{ni}^{R} &= e \int \mathrm{d}^3 x e^{i(\vec{p}_1 - \vec{p}_2 - \vec{p}_3) \cdot \vec{x}} \\
&= e (2\pi)^3 \, \delta^3 (\vec{p}_1 - \vec{p}_2 - \vec{p}_3)
\end{aligned} \tag{130}
$$

  - This is similar to other matrix elements
- First order:

$$T_{fi} = \int \mathrm{d}^3 \vec{p}_\gamma (2\pi)^3 \, \delta^3 \left(\vec{p}_1 - \vec{p}_2 - \vec{p}_\gamma \right) (2\pi)^3 \, \delta^3 \left(\vec{p}_2 - \vec{p}_4 + \vec{p}_\gamma \right) \frac{e^2}{E_i - E_n} \tag{131}$$

  - Energy is not conserved in such intermediate states (otherwise $E_i = E_n$ would blow u

2.4.3.3. Energy Calculations

- We want to calculate the energy of the states

Denotations

- Momenta of particles are

$$\vec{p}_1, \vec{p}_2, \vec{p}_3, \vec{p}_4, \vec{p}_\gamma \tag{132}$$

- The energy of each respective particle is

2.5. More Information on the S-matrix

- The S-matrix describes the interaction between particles in the system, and as previously seen, it is defined as

$$\langle f|S|i\rangle = \mathbb{1} + (2\pi)^4 \delta^4(p_f - p_i)\mathscr{M} \tag{133}$$

- This S-matrix is the primary goal of most calculations in quantum field theory (as it describes the interactions of the particles)

2.5.1. LSZ Reduction

- The way to obtain the S-matrix from the *Time-Ordered Product* involves the **LSZ Reduction Formula** which is extremely important, and allows for the derivation of the extremely simple momentum-space Feynman Rules, which you will see later.
- The LSZ reduction formula is as follows:

$$\begin{aligned}\langle f|S|i\rangle = \left[i\int d^4x_1 e^{-ip_1x_1}(\Box + m^2)\right]\cdots\left[i\int d^4x_n e^{ip_nx_n}(\Box + m^2)\right] \\ \times \langle\Omega|T\phi(x_1)\phi(x_2)\phi(x_3)...\phi(x_n)|\Omega\rangle\end{aligned} \tag{134}$$

Definition 2.5.1.1 (LSZ Reduction): The *LSZ Reduction* is a formula providing S-matrix elements from the time ordered product (in the ground state for the interacting theory Ω).

- Additionally, this formula can allow for the calculation of transfer matrix elements $\mathscr{M}$, but it is more convenient to utilize Feynman diagrams derived from this formulation.
  - The cross sections are calculatable from this as well
- This formula effectively deletes terms in the time ordered product except for those with poles of $\frac{1}{p^2-m^2}$
- The ground state in the interacting theory, $|\Omega\rangle$ is a very inconvenient way to calculate, due to lack of knowledge about the ground state

2.5.1.1. Correlation Functions

- A **correlation function** $\langle\Omega|T\phi(x_1)\cdots\phi(x_n)|\Omega\rangle$ is the time ordered product of fields which appears in the LSZ reduction formula after the integrals
- This is specifically used in the derivation of the LSZ reduction formula

2.5.1.2. Derivation of LSZ Reduction

- Suppose that the initial state $|i\rangle$ and the final state $|f\rangle$ are of the form

$$|i\rangle = \sqrt{2\omega_1}\sqrt{2\omega_2}a^\dagger_{p_1}(-\infty)a^\dagger_{p_2}(\infty)|\Omega\rangle \tag{135}$$

and

$$|f\rangle = \sqrt{2\omega_1}\cdots\sqrt{2\omega_j}a^\dagger_{p_3}(\infty)\cdots a^\dagger_{p_j}(\infty)|\Omega\rangle \tag{136}$$

- Additionally there is the constraint that $|i\rangle \neq |f\rangle$ due to scattering happening.
- The S-matrix element is then

$$\langle f|S|i\rangle = 2^{\frac{j}{2}}\sqrt{\omega_1\cdots\omega_n} \times \langle\Omega|a_{p_3}(\infty)\cdots a_{p_j}(\infty)a^\dagger_{p_1}(-\infty)a^\dagger_{p_2}(-\infty)|\Omega\rangle \tag{137}$$

- Now it is desirable to convert this to scalar fields, ϕ
- From quantization, they are defined by

$$\phi = \int \frac{\mathrm{d}^3p}{(2\pi)^3}\left(a_p(t)e^{-ipx} + a^\dagger_p(t)e^{ipx}\right) \tag{138}$$

2.5.1.2.1. Heisenberg picture to obtain operators

- Consider the Heisenberg picture
  - $a_p(t) = e^{iH(t-t_0)}a_p(t_0)e^{-iH(t-t_0)}$ and $\phi(x) = e^{iH(t-t_0)}\phi(\boldsymbol{x},t_0)e^{-iH(t-t_0)}$ via unitary time evolution
  - The only useful thing to know about $\phi(x,t)$ and $a_p(t)$ is

$$\langle\Omega|\phi(\boldsymbol{x}, t = \pm\infty)|p\rangle = Ce^{-i\boldsymbol{p}\cdot\boldsymbol{x}} \tag{139}$$

- A useful identity:

$$i\int \mathrm{d}^4x e^{ipx}(\Box + m^2)\phi(x) = \sqrt{2\omega_p}\left[a_p(\infty) - a_p(-\infty)\right] \tag{140}$$

  - Proof:

$$\begin{aligned} i\int \mathrm{d}^4x(\Box + m^2)\phi(x) &= i\int \mathrm{d}^4x e^{ipx}\left(\partial_t^2 - \vec{\partial}_x^2 + m^2\right)\phi \\ &= i\int \mathrm{d}^4x e^{ipx}(\partial_t^2 + \vec{p}^2 + m^2)\phi(x) \\ &= i\int \mathrm{d}^4 e^{ipx}(\partial_t^2 + \omega_p^2)\phi(x) \end{aligned} \tag{141}$$

    - Identity for proof:

$$\begin{aligned} &\partial_t\left[e^{ipx}\left(i\partial_t + \omega_p\right)\phi(x)\right] \\ &= \left[i\omega_p e^{ipx}\left(i\partial_t + \omega_p\right) + e^{ipx}\left(i\partial_t^2 + \omega_p\partial_t^2\right)\right]\phi(x) \\ &= ie^{ipx}(\partial_t^2 + \omega^2)\phi(x) \end{aligned} \tag{142}$$

  - Putting this into the integral,

$$\begin{aligned} i\int \mathrm{d}^4x e^{ipx}(\Box + m^2)\phi(x) &= \int \mathrm{d}^4x\partial_t\left[e^{ipx}\left(i\partial_t + \omega_p\right)\phi(x)\right] \\ &= \int \mathrm{d}t\partial_t\left[e^{i\omega_p t}\int \mathrm{d}^3x e^{-i\boldsymbol{p}\cdot\boldsymbol{x}}\left(i\partial_t + \omega_p\right)\phi(x)\right] \end{aligned} \tag{143}$$

- This is true for all $\phi(x)$
- Quantum Field Case:

$$\begin{aligned}
&\int \mathrm{d}^3 x e^{-i\boldsymbol{p}\cdot\boldsymbol{x}}(i\partial_t+\omega_p)\phi(x) \\
&= \int \mathrm{d}^3 x e^{-i\boldsymbol{p}\cdot\boldsymbol{x}}(i\partial_t+\omega_p) \\
&\times \int \frac{\mathrm{d}^3 k}{(2\pi)^3}\frac{1}{\sqrt{2\omega_k}}\left(a_k(t)e^{-ikx}a_k^\dagger(t)e^{ikx}\right) \\
&= \int \frac{\mathrm{d}^3 k}{(2\pi)^3}\int \mathrm{d}^3 x\left(\left(\frac{\omega_k+\omega_p}{\sqrt{2\omega_k}}\right)a_k(t)e^{-ikx}e^{-i\boldsymbol{p}\cdot\boldsymbol{x}}\right. \\
&\left.+\left(\frac{-\omega_k+\omega_p}{\sqrt{2\omega_k}}\right)a_k^\dagger(t)e^{ikx}e^{-i\boldsymbol{p}\cdot\boldsymbol{x}}\right)
\end{aligned} \tag{144}$$

Note (Math Note):
- $\partial_t a_k(t) = 0$ at $t = \pm\infty$

- The $\mathrm{d}^3 x$ integral enforces $\omega_k = \omega_p$ via delta functions
- Therefore, we get

$$\int \mathrm{d}^3 x e^{-i\boldsymbol{p}\cdot\boldsymbol{x}}(i\partial_t+\omega_p)\phi(x) = \sqrt{2\omega_p}a_p(t)e^{-i\omega_p t} \tag{145}$$

- This implies

$$\begin{aligned}
i\int \mathrm{d}^4 x e^{ipx}(\Box+m^2)\phi(x) &= \int \mathrm{d}t\partial_t\left(\left(e^{i\omega_p t}\right)\left(\sqrt{2\omega_p}a_p(t)e^{-i\omega_p t}\right)\right) \\
&= \sqrt{2\omega_p}\left[a_p(\infty)-a_p(-\infty)\right]
\end{aligned} \tag{146}$$

as desired

2.5.1.2.2. S matrix element computing

- Now we can compute

$$\begin{aligned}
\langle f|S|i\rangle &= 2^{\frac{j}{2}}\sqrt{\omega_1\omega_2\ldots\omega_j} \\
&\times \langle\Omega|a_{p_3}(\infty)\ldots a_{p_j}(\infty)a_{p_1}^\dagger(-\infty)a_{p_2}^\dagger(-\infty)|\Omega\rangle
\end{aligned} \tag{147}$$

Definition 2.5.1.2.2.1 (Time Ordered Product):
- The time ordered product $T\mathcal{O}(x_1)\mathcal{O}(x_2)\ldots\mathcal{O}(x_n)$ picks up the operators $\mathcal{O}$ and literally places them so that they are correctly ordered in time. This is the only thing that the operator does, as it keeps the products there.

- Notice that

$$
\begin{aligned}
\langle f|S|i\rangle = \sqrt{2^j \omega_1 \ldots \omega_j} \langle\Omega| T \left[a_{p_3}(\infty) - a_{p_3}(-\infty)\right] \cdots \left[a_{p_j}(\infty) - a_{p_j}(-\infty)\right] \\
\times \left[a^\dagger_{p_1}(-\infty) - a^\dagger_{p_1}(\infty)\right] \left[a^\dagger_{p_2}(-\infty) - a^\dagger_{p_2}(\infty)\right] |\Omega\rangle
\end{aligned}
\tag{148}
$$

- Therefore, we have the LSZ Reduction Formula

$$
\begin{aligned}
\langle p_3 \cdots p_j | S | p_1 p_2 \rangle = \left[i \int \mathrm{d}^4 x_1 e^{-i p_1 x_1} (\Box_1 + m^2) \right] \\
\times \cdots \times \left[i \int \mathrm{d}^4 x_j e^{i p_j x_j} (\Box_j + m^2) \right] \\
\times \langle\Omega| T \phi(x_1) \phi(x_2) \ldots \phi(x_j) |\Omega\rangle
\end{aligned}
\tag{149}
$$

2.5.2. Feynman Propagators

- We can define the feynman propagator $G_F(x, y)$ to be implicitly defined by

$$
(\Box_x + m^2) G_F(x, y) = -i \delta^4(x - y) \tag{150}
$$

- This provides us, in posistion space, with

$$
\begin{aligned}
G_F(x, y) &= \int \frac{\mathrm{d}^4 p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\varepsilon} e^{i p \cdot (x - y)} \\
&\equiv \int \frac{\mathrm{d}^4 p}{(2\pi)^4} G_F(p) e^{-i p x \dot{-} y}
\end{aligned}
\tag{151}
$$

  - The momentum space Feynman Propagator $G_F(p)$ is defined as

$$
G_F(p) \equiv \frac{i}{p^2 - m^2 + i\varepsilon} \tag{152}
$$

2.6. Feynman Diagrams

Definition 2.6.1 (Feynman Diagram): A Feynman diagram is something that is used to represent particle interactions, or simplifications of them which are summed. generally to calculate the LSZ reduction for matrix elements. These matrix elements are ultimately used to calculate cross sections as seen in *Cross Sections*.

Definition 2.6.2 (Feynman Rules): Feynman rules are a set of rules for a theory which is used to evaluate Feynman diagrams.

- A standard set of Feynman rules that can generally apply everywhere are the following

General Position Space Feynman Rules

- At each vertex, a factor of the interaction coupling is given

- All lines are integrated over d^3x after applying the propagator

General Momentum Space Feynman Rules (Arguably much more useful)

- Each vertex of a Feynman Diagram gets a factor of whatever the interaction coupling is.
- Vertices can only occur at points that satisfy interactions in the Lagrangian[6]
- For each edge whose momentum is known (that does not connect to an external vertex), a factor of the propagator for the theory Π is gained
- For each edge whose momentum k is unknown (i.e., a loop), a factor of Π is gained and must be integrated over

$$\frac{\mathrm{d}^4 k}{(2\pi)^4}. \tag{153}$$

- The final product gives $i\mathcal{M}$ for the diagram

Note (Schwinger-Dyson Equation):

[6]For instance, if $\mathcal{L}_{\text{int}} = \frac{g}{3!}\phi^3$, only interactions which involve fields intersecting at three points are included in the set of tree-level diagram

Part III

Spinors and Lots of Math

3.1. Lorentz and Spinors

Some Supplementary material for this section is Spinors for Beginners by Eigenchris which is very useful to get a grasp of the actual math that is going on.

3.1.1. Lorentz Transformations

Definition 3.1.1.1 (Metric Tensor): A (or in special relativity, *The*) metric tensor is defined as being the dot product of basis vectors

$$g_{\mu\nu} = \hat{e}_{(\mu)} \cdot \hat{e}_{(\nu)} \tag{154}$$

This definition is all but useless in Quantum Field Theory due to the whole theory entirely using Special Relativity instead of General Relativity[7]

The special relativity metric tensor (which is subject to arguments over convention), is either $g_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ or $g_{\mu\nu} = \mathrm{diag}(-,+,+,+)$[8].

Definition 3.1.1.2 (Lorentz Transformation & Lorentz Invariance): A *Lorentz transformation* Λ is defined as a coordinate transformation

$$\Lambda : (t, x, y, z) \to (t', x', y', z') \tag{155}$$

such that

$$\Lambda^T g \Lambda = g \tag{156}$$

or

$$\Lambda^\mu_\alpha \Lambda^\nu_\beta \eta_{\mu\nu} = \eta_{\alpha\beta} \tag{157}$$

In special relativity (minkowski space), this condition is loosened to

$$s^2 = t^2 - x^2 \tag{158}$$

is conserved but the previous condition is still useful in discussion of group theory and derivation of *Spinors*. Generally, Lorentz transformations are transformations that

3.1.2. Group Theory (cont'd)

Definition 3.1.2.1 (Lorentz Group): This is the set of operations Λ such that

$$\Lambda^T g \Lambda = g \tag{159}$$

- For instance, a Lorentz transformation could follow the 4-vector representation, defined as

[7] Unless string theory, again

[8] Personally I am used to $\mathrm{diag}(+,-,-,-)$.

$$x_\mu \to \Lambda_{\mu\nu} x_\nu \tag{160}$$

- More generally, a field ϕ transforms under a representation of the lorentz group as

$$\phi_{i(x)} \to R(\Lambda)_{ij} \phi_j (\Lambda^{-1} x) \tag{161}$$

(Similar to a matrix transformation)

- The continuous Lorentz transformations that exist are *rotations* and *boosts*, which are the most general possible (continous) Lorentz transformations which can occur as they (+ the discrete Lorentz transformations such as P and T) form the **Lorentz Group**
- Rotations can be represented, again in the 4-vector representation, as rotation matrices. We let $\theta_x, \theta_y, \theta_z$ represent rotations in the x, y, z axes respectively. Then, the rotation matrices can be represent as as

$$\begin{gathered}
\Lambda(\theta_x) = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \cos\theta_x & \sin\theta_x \\ & & -\sin\theta_x & \cos\theta_x \end{pmatrix}, \Lambda(\theta_y) = \begin{pmatrix} 1 & & & \\ & \cos\theta_y & & -\sin\theta_y \\ & & 1 & \\ & \sin\theta_y & & \cos\theta_y \end{pmatrix} \\
\Lambda(\theta_z) = \begin{pmatrix} 1 & & & \\ & \cos\theta_z & \sin\theta_z & \\ & -\sin\theta_z & \cos\theta_z & \\ & & & 1 \end{pmatrix}
\end{gathered} \tag{162}$$

- In a similar manner, boosts can be represented by matrices in the 4-vector representation. Denoting $\beta_x, \beta_y, \beta_z$ as the boost angles (in the hyperbolic trigonometric functions), we have that

$$\begin{gathered}
\Lambda(\beta_x) = \begin{pmatrix} \cosh\beta_x & \sinh\beta_x & & \\ \sinh\beta_x & \cosh\beta_x & & \\ & & 1 & \\ & & & 1 \end{pmatrix}, \Lambda(\beta_y) = \begin{pmatrix} \cosh\beta_y & & \sinh\beta_y & \\ & 1 & & \\ \sinh\beta_y & & \cosh\beta_y & \\ & & & 1 \end{pmatrix} \\
\Lambda(\beta_z) = \begin{pmatrix} \cosh\beta_z & & & \sinh\beta_z \\ & 1 & & \\ & & 1 & \\ \sinh\beta_z & & & \cosh\beta_z \end{pmatrix}
\end{gathered} \tag{163}$$

- Can utilize *infinitesimal* Lorentz transformations in order to extract the group ($O(1, 3)$) (we let β_i parametrize boosts and θ_i parametrize rotations)
  - $$\begin{gathered} \delta x_0 = \beta_i x_i \\ \delta x_i = \beta_i x_o - \varepsilon_{ijk} \theta_j x_k \end{gathered} \tag{164}$$

  or

$$\delta x_\mu = i \left[\sum_{i \in \{1,2,3\}} \theta (J_i)_{\mu\nu} + \beta_i (K_i)_{\mu\nu} \right] x_\nu \tag{165}$$

  - J_i and K_i matrices:

$$J_1 = i \begin{pmatrix} 0 & & & \\ & 0 & & \\ & & 0 & -1 \\ & & 1 & 0 \end{pmatrix}, J_2 = i \begin{pmatrix} 0 & & & \\ & 0 & & 1 \\ & & 0 & \\ & -1 & & 0 \end{pmatrix}, J_3 = i \begin{pmatrix} 0 & & & \\ & 0 & -1 & \\ & 1 & 0 & \\ & & & 0 \end{pmatrix}$$
$$K_1 = i \begin{pmatrix} 0 & -1 & & \\ -1 & 0 & & \\ & & 0 & \\ & & & 0 \end{pmatrix}, K_2 = i \begin{pmatrix} 0 & & -1 & \\ & 0 & & \\ -1 & & 0 & \\ & & & 0 \end{pmatrix}, K_3 = i \begin{pmatrix} 0 & & & -1 \\ & 0 & & \\ & & 0 & \\ -1 & & & 0 \end{pmatrix} \tag{166}$$

- Such matrices are *generators of the Lorentz group*
  - Can use to obtain the Lie Algebra
- $$\Lambda = \exp(i\theta_i J_i + i\beta_i K_i) \tag{167}$$

3.2. Dirac Spinors

3.2.1. Spinors

Definition 3.2.1.1 (Dirac Spinor): An object ψ obeying the transormation properties

$$\psi \to S[\Lambda]\psi, \psi^\dagger \to \psi^\dagger S[\Lambda] \tag{168}$$

- What does this mean?
- A lorentz transformation can be written as

$$\Lambda^\mu_\nu = \exp\left(\frac{1}{2}\Omega_{\rho\sigma}(J^{\rho\sigma})^\mu_\nu\right) \tag{169}$$

with new notation introduced for convenience such that $\Omega_{0i} = \beta_i$ boosts, $\Omega_{12} = -\Omega_{12} = \theta_z$ (rotation angles) with $(J^{\rho\sigma})^{\mu\nu} = g^{\rho\mu}g^{\sigma\nu} - g^{\sigma\mu}g^{\rho\nu}$

- These are another way to write generators for the lorentz group
- Notice that, using this tensor, we can write

$$K_i = -i(J^{0i})^\mu_\nu \tag{170}$$

for boosts and

$$J_i = \frac{i}{2}\varepsilon_{ijk}(J^{jk})^\mu_\nu \tag{171}$$

for rotations

- These generators form $\mathfrak{so}(1,3)$, the Lorentz Algebra
  - Their exponential gives a representation of the Lorentz Group double cover $\mathrm{Spin}(1,3) \cong \mathrm{SL}(2,\mathbb{C})$

Theorem 3.2.1.1: The Lorentz Algebra, $\mathfrak{so}(1,3)$, has two commuting sub-algebras represented by the 3d rotation algebras $\mathfrak{su}(2)$

Proof:

- Notice that if we switch back to J_i and K_i, we can derive $\mathrm{SU}(2)$, the group for spinors.
- Define $J_i^+ \equiv \frac{1}{2}(J_i + iK_i)$, $J_i^- \equiv \frac{1}{2}(J_i - iK_i)$
- We have then,

$$\begin{aligned} \left[J_i^+, J_j^+\right] &= i\varepsilon_{ijk} J_k^+ \\ \left[J_i^-, J_j^-\right] &= i\varepsilon_{ijk} J_k^- \\ \left[J_i^+, J_i^-\right] &= 0 \end{aligned} \tag{172}$$

  hence, $\mathfrak{so}(1,3)$ has 2 communiting sub algebras, both of which are 3d rotation groups given by the Levi-Cevita connections. Hence, $\mathrm{SO}(1,3) \cong \mathrm{SU}(2) \oplus \mathrm{SU}(2)$

∎

- Hence, we now have the definition of a spinor

Definition 3.2.1.2 (Spinor): A *Spinor* is an object that transforms under $\mathrm{SU}(2)$

3.2.2. Clifford Algebruh

- To construct Dirac Spinors, one must first construct gamma matrices, which form the basis of a Clifford Algebra $\mathscr{C}\ell_{1,3}(\mathbb{R})$

Definition 3.2.2.1 (Gamma Matrices): Gamma matrices are defined as being anticommuting matrices such that

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} \tag{173}$$

and having the property that

$$\left(\gamma^0\right)^2 = 1, \left(\gamma^i\right)^2 = -1 \tag{174}$$

- As seen in the *math toolkit*, these matrices can be represented as

$$\gamma^\mu = \begin{pmatrix} & \sigma^\mu \\ \bar{\sigma}^\mu & \end{pmatrix} \tag{175}$$

  with $\sigma^\mu = (\mathbb{1}, \vec{\sigma}), \bar{\sigma}^\mu = (\mathbb{1}, -\vec{\sigma})$ and σ^i being Pauli Matrices
  - This representation is known as the Weyl Representation
- An additional piece of useful notation is the idea that $\not{p} = p_\mu \gamma^\mu = p^\mu \gamma_\mu$

Pauli Matrix and Gamma Matrix Indentities

$$\sigma^i\sigma^j = \delta^{ij} i\varepsilon^{ijk}\sigma^k, \{\sigma^i, \sigma^j\} = 2\delta^{ij}, [\sigma^i, \sigma^j] = 2i\varepsilon^{ijk}\sigma^k \tag{176}$$

For Pauli Matrices

- A useful quantity to define is a representation of the Lorentz Algebra

$$S^{\rho\sigma} = \frac{1}{2}[\gamma^\rho, \gamma^\sigma] = \frac{1}{2}\gamma^\rho\gamma^\sigma - \frac{1}{2}g^{\rho\sigma} \tag{177}$$

- The reason that this is a representation of the

Part IV

QED

Definition 4.1 (Quantum Electrodynamics): Quantum Electrodynamics is a relatively simple theory with only one force involved (as suggested by the name, it is the electromagnetic force), mediated by the *photon* A_μ and one/two type(s) of fermion depending on what you consider a type of particle (an *electron*/*positron*)

4.1. Scalar QED

- Lagrangian is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^2 + \left|D_\mu \phi\right|^2 - m^2 \phi^* \phi \tag{178}$$

  - $D_\mu \phi = \partial_\mu \phi + ieA_\mu$
  - $\left(D_\mu \phi\right)^* = \partial_\mu \phi^* - iA_u$

4.1.1. Charge

- The charge in quantum electrodynamics is associated with the $\mathrm{U}(1)$ symmetry

$$\phi \to e^{-i\alpha(x)}\phi \tag{179}$$

  (in fact, this applies in spinor QED as well, due to the)
-

4.2. Spinor QED (Standard Quantum Electrodynamics)

- The Spinors in Quantum Electrodynamics are Dirac Spinors due to their presence of charge (Since Majorana spinors are the same as their antiparticle, they of course cannot be a candidate for this, and hence the spinors must be Dirac Spinors).

4.2.1. Solution to the Dirac Equation

- Similarly to a Scalar field theory, we will quantize particles by writing

$$\psi_s(x) = \int \frac{\mathrm{d}^3 p}{(2\pi)^3} u_s(p) e^{ipx} \tag{180}$$

  for particles and

$$\chi_s(x) = \int \frac{\mathrm{d}^3 p}{(2\pi)^3} v_s(p) e^{ipx} \tag{181}$$

  for antiparticles
- These $u_s(p)$ and $v_s(p)$ are *basis spinors* which satisfy the momentum space dirac equation. That is,

$$\begin{pmatrix} -m & p \cdot \sigma \\ p \cdot \bar{\sigma} & -m \end{pmatrix} u_s(p) = \begin{pmatrix} -m & -p \cdot \sigma \\ -p \cdot \bar{\sigma} & -m \end{pmatrix} v_s(p) = 0 \tag{182}$$

- Again, these $u_s(p)$ and $v_s(p)$ are 4-index Dirac spinors.

Basis Dirac Spinors

In general, the solution to this equation is

$$u_s(p) = \begin{pmatrix} \sqrt{p \cdot \sigma}\xi \\ \sqrt{p \cdot \bar{\sigma}}\xi \end{pmatrix}, v_s(p) = \begin{pmatrix} \sqrt{p \cdot \sigma}\eta_s \\ -\sqrt{p \cdot \bar{\sigma}}\eta_s \end{pmatrix} \tag{183}$$

- $$\sum_s u_s(p)\bar{u}_s(p) = \not{p} + m \tag{184}$$
- $$\sum_s v_s(p)\bar{v}_s(p) = \not{p} - m \tag{185}$$

4.2.2. CPT Theorem

- In order to have the entire Lorentz Group, we must define the following discrete transforms for spinors.
- Time reversal T which maps
$$T : (t, \vec{x}) \to (-t, \vec{x}) \tag{186}$$
- Parity P which maps
$$P : (t, \vec{x}) \to (t, -\vec{x}) \tag{187}$$
- Charge conjugation
$$C : \psi \to -i\gamma^2\psi \tag{188}$$
reverses charge (essentially it is particle $\to$ antiparticle)

4.2.3. Interacting QED

- The Lagrangian for Quantum Electrodynamics is
$$\begin{aligned} \mathcal{L} &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + i\bar{\psi}\not{D}\psi - m\bar{\psi}\psi \\ &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\gamma^\mu\partial\mu - m)\psi - e\bar{\psi}\gamma^\mu\psi A_\mu \end{aligned} \tag{189}$$
- This gives rise to the interaction $-ie\gamma^\mu$ with the Feynman Diagram

Theorem 4.2.3.1 (Spin-Statistics Theorem): The Spin statistics theorem is the theorem that spinors are governed by Fermi-Dirac statistics and that Vectors are governed by Bose-Einstein statistics. Essentially $\psi(x)\chi(y) = -\chi(x)\psi(y)$ and

- The Feynman propagator for spinors is
$$\langle 0|T\psi(0)\bar{\psi}(x)|0\rangle = \int \frac{\mathrm{d}^4p}{(2\pi)^4} \frac{i(\not{p} + m)}{p^2 - m^2 + i\varepsilon} e^{ipx} \tag{190}$$
with of course the implicit limit of $\varepsilon \to 0$

- The derivation for this is as follows
  -

- The Feynman propagator clearly gives rise to the electron propagator present on internal electrons

$$\text{feynman diagram} = \frac{i(\not{p} + m)}{p^2 - m^2 + i\varepsilon} \tag{191}$$

4.3. Renormalization

- Let us take a look at the 1-loop correction to the free electron.
- Evaluating this loop gives an infinite quantity which we would like to analyze (it is considered a UV Divergence)
- In order to properly evaluate such integrals to give a physical quantity, it is key to utilize a technique known as **renormalization**
  - There are different types of renormalization, such as parametrizing integral bounds, taking the limit of the number of dimensions, etc

Part IVIII

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